

Reaction–Diffusion Exchange Economies with Aggregated Cobb–Douglas Preferences for $n \geq 2$ Goods

Diego Correa Tristain
 algoritmia@labormedia.cl

January 28, 2026

Abstract

This document extends the two-good, two-stage convex zero-sum exchange model introduced in *Emergent Properties of Distributed Agents with Two-Stage Convex Zero-Sum Optimal Exchange Network* (Tristain 2024) to an economy with $n \geq 2$ goods. The extension is made possible by aggregating *pairwise* Cobb–Douglas preference parameters into a globally consistent Cobb–Douglas exponent vector $\beta^{(i)}$ for each agent i . The aggregation is exact when a cycle-consistency condition holds; otherwise it can be posed as a least-squares projection in log-ratio space. The resulting n -good exchange economy preserves convexity and tractability, and supports reaction–diffusion dynamics in the sense of Turing-type pattern formation (Turing 1952).

1 Introduction

We consider an economy with:

- $m \geq 2$ agents indexed by $i \in \{1, \dots, m\}$,
- $n \geq 2$ goods (“species”) indexed by $a \in \{1, \dots, n\}$,
- discrete time (rounds) indexed by $t \in \{0, 1, 2, \dots\}$.

Each agent i holds an endowment vector

$$\mathbf{e}_i^t = (e_{i,1}^t, \dots, e_{i,n}^t)^\top \in \mathbb{R}_+^n,$$

and the system satisfies a global conservation law across agents in the *diffusion (exchange) phase*:

$$\sum_{i=1}^m \mathbf{e}_i^t = \mathbf{E} \in \mathbb{R}_+^n \quad \forall t.$$

In the original two-good model, bilateral exchanges were written as a zero-sum constraint at the edge level. For n goods, a bilateral exchange between agents i and j can be expressed as an exchange vector $\boldsymbol{\varepsilon}_{i \rightarrow j}^t \in \mathbb{R}^n$ such that

$$\boldsymbol{\varepsilon}_{i \rightarrow j}^t + \boldsymbol{\varepsilon}_{j \rightarrow i}^t = \mathbf{0},$$

ensuring that trades only redistribute goods among agents.

The dynamics have two phases each round:

1. **Reaction:** each agent applies endogenous transformation rules (technologies) to its own endowment, mapping some goods into others under feasibility constraints.
2. **Diffusion:** agents exchange goods via local, utility-maximizing trades subject to budgets, producing a redistribution consistent with conservation.

This mirrors reaction–diffusion systems in which reaction reshapes local composition while diffusion transports quantities across the network (Turing 1952). The key additional ingredient in the economic setting is that diffusion is not random transport but *Pareto-improving exchange* governed by convex preferences.

2 Pairwise Cobb–Douglas Preferences

Fix an agent i . For each ordered pair of *distinct* goods (a, b) the agent specifies a pairwise Cobb–Douglas utility over the two-dimensional projection (x_a, x_b) :

$$u_{i,(a,b)}(x_a, x_b) = x_a^{\alpha_{ab}^{(i)}} x_b^{1-\alpha_{ab}^{(i)}}, \quad \alpha_{ab}^{(i)} \in (0, 1).$$

A natural symmetry requirement is

$$\alpha_{ba}^{(i)} = 1 - \alpha_{ab}^{(i)},$$

so that the same indifference curves are obtained regardless of the direction in which the pair is written.

2.1 Preference ratios

Define the pairwise *intensity ratio* (good b relative to a):

$$r_{ab}^{(i)} := \frac{1 - \alpha_{ab}^{(i)}}{\alpha_{ab}^{(i)}} > 0.$$

In a global Cobb–Douglas representation, these ratios correspond to exponent ratios:

$$r_{ab}^{(i)} = \frac{\beta_b^{(i)}}{\beta_a^{(i)}}.$$

3 Aggregation to a global Cobb–Douglas utility

We seek a global utility for agent i of the form

$$u_i(\mathbf{x}) = \prod_{a=1}^n x_a^{\beta_a^{(i)}}, \quad \mathbf{x} \in \mathbb{R}_{++}^n, \quad \beta_a^{(i)} > 0,$$

where

$$\boldsymbol{\beta}^{(i)} := (\beta_1^{(i)}, \dots, \beta_n^{(i)})^\top \in \mathbb{R}_{++}^n$$

is the (unnormalized) exponent vector. Normalizing to unit sum gives

$$\tilde{\boldsymbol{\beta}}^{(i)} := \frac{\boldsymbol{\beta}^{(i)}}{\mathbf{1}^\top \boldsymbol{\beta}^{(i)}}, \quad \mathbf{1} \in \mathbb{R}^n,$$

so that $\sum_a \tilde{\beta}_a^{(i)} = 1$.

3.1 Exact consistency

The pairwise utilities are exactly consistent with some $\beta^{(i)}$ if and only if for all distinct a, b

$$\alpha_{ab}^{(i)} = \frac{\beta_a^{(i)}}{\beta_a^{(i)} + \beta_b^{(i)}} \iff \frac{\beta_b^{(i)}}{\beta_a^{(i)}} = r_{ab}^{(i)}.$$

Equivalently, every cycle $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_k \rightarrow a_1$ must satisfy the cycle condition:

$$\prod_{s=1}^k r_{a_s a_{s+1}}^{(i)} = 1, \quad (a_{k+1} = a_1).$$

This is the multiplicative “no-arbitrage” condition in ratio space, ruling out intransitive preference cycles.

3.2 Constructing $\beta^{(i)}$

If the ratios are consistent, $\beta^{(i)}$ is determined up to a positive scalar. Fix a reference good (e.g. good 1) and set $\beta_1^{(i)} = 1$. Then for each a , choose any path $1 \rightarrow \dots \rightarrow a$ and propagate:

$$\beta_a^{(i)} = \left(\prod_{\ell \text{ on path}} r_{\ell, \ell+1}^{(i)} \right) \beta_1^{(i)}.$$

The cycle condition guarantees path-independence. Finally normalize to obtain $\tilde{\beta}^{(i)}$.

3.3 Noisy or inconsistent pairwise inputs

Empirical $\alpha_{ab}^{(i)}$ will generally be noisy and violate the cycle condition. A convenient projection is to work in log space. Let $y_a^{(i)} := \log \beta_a^{(i)}$ and $\rho_{ab}^{(i)} := \log r_{ab}^{(i)}$. Then exact consistency is

$$y_b^{(i)} - y_a^{(i)} = \rho_{ab}^{(i)} \quad \forall a \neq b,$$

and a least-squares recovery solves

$$\min_{\mathbf{y}^{(i)} \in \mathbb{R}^n} \sum_{1 \leq a < b \leq n} \left((y_b^{(i)} - y_a^{(i)}) - \rho_{ab}^{(i)} \right)^2,$$

with a gauge fixing such as $y_1^{(i)} = 0$. Setting $\beta_a^{(i)} := e^{y_a^{(i)}}$ then yields a best-fit exponent vector, which can be normalized.

4 The n -good reaction–diffusion exchange economy

4.1 Reaction phase (endogenous transformation)

Let $\mathcal{R}$ be a finite set of reaction rules (technologies). Each rule $r \in \mathcal{R}$ is specified by:

- an *input* vector $\mathbf{a}_r \in \mathbb{R}_+^n$,

- an *output* vector $\mathbf{b}_r \in \mathbb{R}_+^n$,
- an intensity (decision variable) $x_{i,r}^t \geq 0$ for agent i at time t .

Agent i 's post-reaction endowment is

$$\mathbf{e}_i^{t,\text{react}} = \mathbf{e}_i^t + \sum_{r \in \mathcal{R}} x_{i,r}^t (\mathbf{b}_r - \mathbf{a}_r),$$

subject to feasibility (componentwise):

$$\sum_{r \in \mathcal{R}} x_{i,r}^t \mathbf{a}_r \leq \mathbf{e}_i^t.$$

This defines a convex feasible set because the constraints are linear and the update is affine in $\{x_{i,r}^t\}$.

Conservation remark. Reaction rules typically change the composition of $\mathbf{E}$ over time. If one wishes to enforce conservation of a linear invariant (e.g. mass/energy), introduce a matrix L such that $L\mathbf{b}_r = L\mathbf{a}_r$ for all r , which guarantees $L \sum_i \mathbf{e}_i^t$ is preserved.

4.2 Diffusion phase (exchange)

Let $\mathbf{p}^t \in \mathbb{R}_{++}^n$ be a price vector at round t . Each agent chooses a *per-good* trade fraction

$$\boldsymbol{\lambda}_i^t = (\lambda_{i,1}^t, \dots, \lambda_{i,n}^t)^\top \in [0, 1]^n,$$

and posts a supply vector (Hadamard product $\odot$):

$$\mathbf{s}_i^t = \boldsymbol{\lambda}_i^t \odot \mathbf{e}_i^{t,\text{react}}.$$

The retained inventory is $(\mathbf{1} - \boldsymbol{\lambda}_i^t) \odot \mathbf{e}_i^{t,\text{react}}$.

Let $\mathbf{z}_i^t \in \mathbb{R}_+^n$ denote the agent's acquired bundle from exchange. Define the trade budget (wealth committed to trading) as

$$w_i^t := \mathbf{p}^t \cdot (\boldsymbol{\lambda}_i^t \odot \mathbf{e}_i^{t,\text{react}}).$$

The diffusion problem is

$$\max_{\boldsymbol{\lambda}_i^t \in [0,1]^n, \mathbf{z}_i^t \geq 0} u_i \left((\mathbf{1} - \boldsymbol{\lambda}_i^t) \odot \mathbf{e}_i^{t,\text{react}} + \mathbf{z}_i^t \right) \quad \text{s.t.} \quad \mathbf{p}^t \cdot \mathbf{z}_i^t \leq w_i^t. \quad (1)$$

4.3 Marshallian demand for fixed $\boldsymbol{\lambda}_i^t$

For fixed $\boldsymbol{\lambda}_i^t$, the problem in $\mathbf{z}_i^t$ is standard Cobb–Douglas with linear budget. Let $\tilde{\boldsymbol{\beta}}^{(i)}$ be the normalized exponents. The Marshallian demand is

$$z_{i,a}^t(\boldsymbol{\lambda}_i^t) = \frac{\tilde{\beta}_a^{(i)}}{p_a^t} w_i^t, \quad a = 1, \dots, n.$$

The post-trade holdings are

$$x_{i,a}^t(\boldsymbol{\lambda}_i^t) = (1 - \lambda_{i,a}^t) e_{i,a}^{t,\text{react}} + z_{i,a}^t(\boldsymbol{\lambda}_i^t).$$

4.4 Concave optimization over λ_i^t

Define the log-utility as a function of λ_i^t :

$$\ell_i(\lambda_i^t) := \log u_i(x_i^t(\lambda_i^t)) = \sum_{a=1}^n \tilde{\beta}_a^{(i)} \log x_{i,a}^t(\lambda_i^t).$$

Since each $x_{i,a}^t(\lambda)$ is affine in λ and log is concave, ℓ_i is concave on $[0, 1]^n$. Hence the optimization in λ_i^t is a convex program (maximize concave objective with box constraints), solvable by projected gradient, coordinate ascent, or off-the-shelf convex solvers.

A useful closed-form expression for the partial derivatives is obtained by writing

$$w_i^t = \sum_{b=1}^n p_b^t \lambda_{i,b}^t e_{i,b}^{t,\text{react}}, \quad x_{i,a}^t = (1 - \lambda_{i,a}^t) e_{i,a}^{t,\text{react}} + \frac{\tilde{\beta}_a^{(i)}}{p_a^t} w_i^t.$$

Then for each component $m \in \{1, \dots, n\}$,

$$\frac{\partial \ell_i}{\partial \lambda_{i,m}^t} = -\tilde{\beta}_m^{(i)} \frac{e_{i,m}^{t,\text{react}}}{x_{i,m}^t} + p_m^t e_{i,m}^{t,\text{react}} \sum_{a=1}^n \tilde{\beta}_a^{(i)} \frac{\tilde{\beta}_a^{(i)}}{p_a^t x_{i,a}^t}.$$

At an interior optimum, these partial derivatives vanish (KKT conditions), while boundary optima occur at $\lambda_{i,m}^t \in \{0, 1\}$ for some components.

5 Market clearing and price determination

Total posted supply of good a is

$$Q_a^t = \sum_{i=1}^m \lambda_{i,a}^t e_{i,a}^{t,\text{react}}.$$

Total demand (from Marshallian demand) is

$$D_a^t = \sum_{i=1}^m \frac{\tilde{\beta}_a^{(i)}}{p_a^t} w_i^t, \quad w_i^t = \mathbf{p}^t \cdot (\lambda_i^t \odot \mathbf{e}_i^{t,\text{react}}).$$

Market clearing requires $D_a^t = Q_a^t$ for all a .

If all agents share the same exponent vector $\tilde{\beta}$ and choose a common scalar trading rate $\lambda_i^t \equiv \lambda^t$, then relative prices recover the familiar closed form

$$\frac{p_a^t}{p_b^t} = \frac{\tilde{\beta}_a Q_b^t}{\tilde{\beta}_b Q_a^t}.$$

In heterogeneous or fully vector-valued λ_i^t settings, prices are naturally computed as a fixed point (e.g. via tâtonnement or an excess-demand root-finder).

6 Conclusion

Aggregating pairwise Cobb–Douglas preferences into a global exponent vector enables a faithful n -good extension of the two-good model in (Tristain 2024). The extended reaction–diffusion economy preserves:

- convexity of feasible reaction sets and diffusion optimization,
- closed-form Marshallian demand conditional on trading rates,
- a principled interpretation of endogenous transformations (reaction) plus Pareto-improving exchange (diffusion),
- and a clear pathway to simulation and implementation for $n \geq 2$ goods.

References

- Boyd, Stephen and Lieven Vandenberghe (2004). *Convex Optimization*. Cambridge University Press. ISBN: 9780521833783. URL: <https://web.stanford.edu/~boyd/cvxbook/>.
- Cobb, Charles W. and Paul H. Douglas (1928). “A Theory of Production”. In: *The American Economic Review* 18.1, pp. 139–165.
- Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green (1995). *Microeconomic Theory*. Oxford University Press. ISBN: 9780195102680.
- Tristain, Diego Correa (2024). *Emergent Properties of Distributed Agents with Two-Stage Convex Zero-Sum Optimal Exchange Network*. Open-access manuscript and repository. URL: https://github.com/onedge-network/Emergent_Properties_paper (visited on 01/28/2026).
- Turing, Alan M. (1952). “The Chemical Basis of Morphogenesis”. In: *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 237.641, pp. 37–72. DOI: 10.1098/rstb.1952.0012.
- Varian, Hal R. (1992). *Microeconomic Analysis*. 3rd ed. W. W. Norton & Company. ISBN: 9780393957358.